

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I NS BHAVESH(192565986) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

NS BHAVESH(192565986)

Signature of the student:

Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.
3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.
5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically

separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

Question 1 — Star Topology for a 20-Person Startup

Problem: Design a cost-effective, easy-to-manage network for a 20-person startup with workstations, printers, and VoIP phones.

Problem Analysis

- A small (20-user) office needs simple management, easy troubleshooting, low upfront cost, and room to grow — ruling out complex mesh designs and pointing directly to a Star topology.
- Key requirements: fault isolation per device, Gigabit-capable wired access for workstations/VoIP, and wireless coverage for laptops/mobiles.

Solution Design — Network Architecture

- Core: one Layer 2/Layer 3 managed switch (e.g., Cisco Catalyst / Netgear ProSafe) with 24+ ports — enough for all workstations, printers, and VoIP phones with headroom for growth.
- Every end device connects individually, in a hub-and-spoke pattern, directly to the central switch — the defining structure of a star topology.
- The switch uplinks to a router/firewall appliance (Cisco RV series or pfSense) that handles internet access, DHCP, and NAT for the whole office.
- A Wi-Fi access point connects to the switch to extend wireless coverage to laptops and mobile devices.

Technical Implementation — Cabling & Devices

- Cat6 UTP cabling runs from the switch to each device — supports Gigabit speeds up to 100 m, more than sufficient for a single office floor.
- Structured, centralized cable management (patch panel at the switch) simplifies future moves, adds, and changes (MACs).

Use of Tools & Technologies

- Managed Layer 2/3 switch for VLAN segmentation and future scalability (e.g., separating VoIP traffic onto its own VLAN).
- SOHO/SMB router-firewall for NAT, DHCP, and basic perimeter security.
- Cat6 UTP structured cabling and a wireless access point for mixed wired/wireless access.

Justification / Advantage

- Fault isolation: if one workstation's cable fails, only that node goes offline — the other 19 employees are unaffected, since every device has its own independent link back to the switch.
- Low complexity and cost make this the right-sized solution for a 20-person startup, compared to the added expense and complexity of a mesh design.

Question 2 — Mesh Topology for a Bank Requiring Near-Zero Downtime

Problem: Design a highly redundant network for a bank where near-zero downtime and regulatory uptime/data-integrity requirements must be met.

Problem Analysis

- Financial transaction systems cannot tolerate single points of failure — any core link or device outage must not disrupt service, and failover must happen in milliseconds.
- This requirement for maximum redundancy at the core directly points to a full or partial Mesh topology rather than star or bus.

Solution Design — Network Architecture

- Full mesh at the core: every core device — routers, core switches, and servers — connects directly to every other core device, so no single link or node failure isolates any part of the network.
- Partial mesh is used practically: core switches/firewalls are fully meshed with each other, while distribution and access layers use redundant uplinks to at least two core switches (rather than meshing every access device, which would be cost-prohibitive).
- Dual enterprise-grade routers (Cisco ASR / Juniper MX) provide BGP-based redundant internet connectivity, so the bank is not dependent on a single ISP link.

Technical Implementation — Cabling & Redundancy

- All core-to-core links use fiber optic cabling: single-mode fiber for long inter-building runs, multi-mode OM4 within the data center — chosen for immunity to electromagnetic interference, very high bandwidth, and the low latency financial transactions require.
- Redundant power supplies, UPS systems, and RAID storage arrays complement the network-level redundancy at the infrastructure level.

Use of Tools & Technologies

- Dynamic routing protocols — OSPF (interior) and BGP (redundant internet edge) — enable automatic, millisecond-scale rerouting around a failed link or node.
- Enterprise-grade routers/switches with hardware redundancy (dual power supplies, hot-swappable modules).

Justification / Advantage

- When a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds via OSPF/BGP, satisfying regulatory requirements for uptime and data integrity — the core reason mesh is the correct topology here.

Question 3 — Bus Topology for a Low-Budget Classroom Lab

Problem: Design the lowest-cost feasible network for a small, supervised classroom lab with light usage (browsing, file sharing).

Problem Analysis

- The primary constraint is minimum hardware cost, with only light, supervised usage — collisions and a single point of cable failure are acceptable trade-offs in this context, which makes Bus topology (or a bus-like low-cost variant) the appropriate fit.

Solution Design — Network Architecture

- Classic implementation: a single coaxial backbone cable (RG-58, 50-ohm) runs the length of the room; all computers share this one common communication line, eliminating the need for a central switch/hub.
- Each computer taps into the backbone via a T-connector and BNC connector; 50-ohm terminators are placed at both ends of the cable to absorb signals and prevent reflection.
- Modern low-cost variant: a single inexpensive 8- or 16-port unmanaged switch replaces the coax/terminators, using Cat5e UTP patch cables to each workstation — achieving bus-like simplicity and cost at minimal expense using modern parts.

Technical Implementation

- No dedicated server is required; a teacher's PC shares resources using built-in Windows/Linux file sharing, keeping costs to just cabling and one inexpensive switch (or coax kit).

Use of Tools & Technologies

- RG-58 coaxial cable with BNC T-connectors and terminators (classic setup), or an unmanaged Ethernet switch with Cat5e UTP (modern low-cost setup).

Limitations Acknowledged

- Susceptible to collisions under heavier load, and a single cable break can disrupt all connected machines — an acceptable trade-off given the supervised, light-usage classroom context, but not suitable for a mission-critical environment.

Question 4 — Hybrid Topology for a Large University Campus

Problem: Design a scalable, resilient network for a large campus with thousands of users spread across lecture halls, labs, libraries, dormitories, and administration buildings.

Problem Analysis

- Scale and diversity of buildings/usage make any single topology inadequate: the backbone needs redundancy, each building needs manageability, and individual labs need cost efficiency — this combination of needs is what a Hybrid topology is designed to solve.

Solution Design — Three-Layer Hierarchical Architecture

- Core layer: a fiber-based ring or partial mesh interconnects core routers and multilayer switches at the campus backbone, so no single link failure isolates an entire building.
- Distribution layer: each building has its own distribution switch, connected to the campus core via dual fiber uplinks for redundancy.
- Access layer: within each building, a star topology branches from the distribution switch to floor-level access switches, which connect individual devices via Cat6 cabling; individual labs may use bus/tree structures for further cost efficiency.

Technical Implementation — Cabling & Devices

- Campus core links use single-mode fiber optic cabling at 10 Gbps or 40 Gbps for high-capacity, long-distance, interference-immune backbone connectivity.
- Wireless access points throughout campus connect via PoE (Power over Ethernet) to access switches, removing the need for separate power cabling to each AP.

Use of Tools & Technologies

- Core/distribution/access three-layer hierarchical design — the industry-standard model for enterprise and campus networks.
- Multilayer switches and core routers running ring/mesh redundancy protocols at the backbone.

Justification / Scalability

- Adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure — demonstrating the design's scalability.
- The hybrid design delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide — addressing all three competing requirements simultaneously.

Question 5 — Multi-Department Company Network (VLAN-Segmented Hierarchical Design)

Problem: Design a structured, scalable, and secure network for a company with multiple departments (HR, Finance, Sales, IT).

Problem Analysis

- Each department needs internal communication among its own devices, controlled communication with other departments, and protection from unnecessary broadcast traffic and security exposure — requirements best met with a VLAN-segmented, hierarchical switch/router design.

Solution Design — Network Architecture

- Access layer: each department (HR, Finance, Sales, IT) connects through its own dedicated Layer 2 switches, allowing computers and printers within that department to communicate locally.
- Core layer: all department switches connect up to a central Layer 3 switch/router, which enables inter-VLAN routing — controlled communication between departments.
- VLANs are configured to logically separate each department's traffic, improving security (departments cannot see each other's broadcast/local traffic by default) and reducing overall broadcast traffic on the network.
- A router with firewall functionality connects the internal network to the internet/external networks, providing perimeter security.

Technical Implementation

- Proper IP addressing and subnetting is applied per department/VLAN for efficient, organized network management.
- Redundant links and backup paths are included between the access and core layers to improve reliability.

Use of Tools & Technologies

- Layer 2 access switches per department, a Layer 3 core switch/router for inter-VLAN routing, and a firewall-capable router/appliance for internet connectivity.
- VLAN tagging (802.1Q) to logically segment departmental traffic over shared physical infrastructure.

Justification / Advantage

- This hierarchical, VLAN-segmented design improves performance (reduced broadcast domains), enhances security (logical isolation between departments), and allows easy expansion as the company grows — new departments simply get a new VLAN and access switch port(s).